Enhanced What-If Scenarios Generation by Bridging Generative Models and Process Simulation

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***Abstract*—Generative Artificial Intelligence (GenAI) techniques have recently emerged as promising tools for challenges in Process Mining such as missing data, small datasets, and the generation of complex what-if scenarios. A major limitation of current approaches is their inability to produce sets of process traces in a mutually dependent manner. Traces are generated independently, failing to reflect crucial process-level phenomena such as concurrency, resource contention, and workload distribution. This limitation significantly reduces the realism and utility of generated logs, yet has received limited attention in the literature.**

In contrast, Business Process Simulation (BPS) excels at preserving process constraints and interdependencies. However, in the presence of specific or complex what-if scenarios, this requires the adjustment of several simulation parameters, which can be challenging to achieve.

Motivated by these issues, where generative methods lack consistency and BPS methods lack flexibility, we propose a novel data generation framework combining GenAI with BPS. The proposed approach enables generating diverse and complex what-if scenarios via data-driven generative models, which are then validated and refined through BPS to ensure consistency and feasibility at the log level. We call the approach What-If Scenarios via Data-driven generative mOdels and siMulation (WISDOM).

We compare several existing methods for process data generation and assess their outputs using an established evaluation framework. Results show that while generative methods fail to capture the full intricacies of process behaviour, their integration with BPS produces traces that are both diverse and compliant with process constraints. This hybrid approach addresses a critical gap in the literature and opens new avenues for robust, simulation-grounded generation methods in process mining.

***Index Terms*—generative AI, what-if scenarios, business process simulation**

I. Introduction

Process mining (PM) [1] is a research domain dedicated to analysing, monitoring, and enhancing business processes using event log data. Multiple generative Artificial Intelligence (GenAI) techniques have been proposed in the Process Mining literature as powerful tools for producing synthetic event traces that mimic the behaviour of real-world processes [2]. According to [3], generative models for Process Mining can be categorized into two main families: GenAI and Business Process Simulation (BPS) techniques. *GenAI models* have been proposed for a large range of applications, such as predictive monitoring [4], what-if scenario analysis [5], counterfactual generation [6], and conditional data generation [7]. While these techniques have shown promising results in generating traces that follow different distributions from the training data, they exhibit two critical flaws: (i) they cannot generalize beyond the behaviours observed in the training data, and (ii) they generate single, independent traces that fail to reflect inter-trace dependencies present in real-world processes.

On the other hand, BPS has seen rapid advancements in recent years. Due to their explicit representation of process dynamics and resource constraints, BPS models are naturally suited to generate traces simultaneously and capture inter-trace dependencies such as resource contention, queuing effects, and process congestion. They are also often regarded as a well-suited tool for conducting what-if scenario analysis given their transparent structure [5]. However, this potential has so far been only partially explored, and mainly limited to relatively simple scenarios, such as resource management [8] or process capacity analysis [9]. Even so, BPS techniques struggle with more complex what-if scenarios, which are challenging to achieve, especially when multiple parameters must be carefully tuned.

Therefore, in this work we leverage the respective strengths of these two families of approaches; on the one hand, GenAI models excel at generating samples that capture an exhaustive representation of the process behaviour, but can only do so in an independent way, while on the other hand, BPS techniques excel at handling inter-trace dependencies and ensuring the compliance of resource contention. Through this intersection, we propose WISDOM (What-If Scenarios via Data-driven generative mOdels and siMulation), a process data generation

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framework that combines generative models with BPS techniques to produce diverse what-if scenarios while ensuring adherence to all process constraints.

In the evaluation, we first show that while GenAI techniques can generate high-quality independent traces, they struggle to maintain inter-trace dependencies, demonstrating the need for an integrated generative approach with BPS. Secondly, we show that the integration within WISDOM maintains the quality of the traces while solving the aforementioned issue. We employ an evaluation protocol to determine the quality of the generated data, such as the accuracy of the control flow, time, and resource perspectives [10].

The rest of this article is structured as follows: Section II presents the motivating example, Section III provides the necessary background concepts, while Section IV discusses related works. The WISDOM approach is introduced in Section V, followed by the evaluation protocol in Section VI. The overall results are detailed in Section VII, while conclusions and directions for future work presented in Section VIII.

II. Motivating Example

In this section, we present a motivating example that highlights the advantages of integrating BPS with generative models. Consider the BPMN model in Figure 1, which represents the loan application process inspired by the BPI Challenge 2017 [11]. A loan application can be created through an initial submission on the website by the client (W_Handle leads) followed by refinement by an employee, or can be directly submitted by the employee (W_Complete application). A shortened procedure is available for clients with a lower credit risk profile (W_Shortened completion). The process continues with the generation of loan offers, which includes activities such as W_Call after offer and W_Validate application, or it ends if the clients decide to abandon the submitted application. Additionally, the bank may request that necessary documents be re-sent to finalize the offer (W_Call incomplete files). A fraud evaluation (W_asses potential fraud) may also be conducted as part of the application review. Each loan application is characterized by three attributes that may affect the progress and the outcome of the application: the requested loan amount (*RequestedAmount*), the reason for the loan application (*LoanGoal*), such as Home, Car, or Business, and the type of application, whether it is a limit increase or a new credit request (*ApplicationType*).

Let us assume that our goal is to support the bank in making strategic decisions by evaluating the feasibility and benefits of potential process changes. Specifically, the bank intends to simplify the procedure for requesting loan limit increases for car purchases and evaluate how this change affects the overall process. Previously applied to 2% of applications, the procedure is now expected to cover roughly 10%. To evaluate this potential increase, we aim to generate a what-if scenario by adding new instances that follow the new procedure, while preserving the behaviours of the original applications.

A recent state-of-the-art solution to support the bank's decision-making is provided by GenAI models, which are able to generate plausible traces representing the desired what-if scenario by leveraging and reusing statistical patterns learned from the entire set of historical traces [12]–[14]. Such methods generate a set of instances that satisfy the conditions and constraints associated with the desired what-if scenario [12]. Nonetheless, this independent construction of traces does not guarantee the preservation of correlations and compliance. In particular, when resources are shared across traces, GenAI approaches provide a distorted estimation of resource utilization and timing, leading to incorrect strategic decisions.

On the other hand, BPS is a well-established and effective technique for defining what-if scenarios [5], and it can overcome the issue of interdependencies between traces during simulation. Starting from a simulation model—either discovered from data or defined by domain experts—we can leverage it to construct our what-if scenario. However, the more complex the process, the more sophisticated the simulation model needs to be, particularly in terms of the simulation parameters. These parameters govern the execution of simulated traces over the process model (as the BPMN model in Figure 1) and relate to various process perspectives, such as the resource and time perspectives. This in turn makes defining what-if scenarios with BPS particularly challenging, as it requires modifying multiple simulation parameters or may lead to inaccurate results if the parameters are overly simplified. Indeed, creating the what-if scenario for the new procedure of loan process involves modifying several simulation parameters: excluding the activity W_Handle leads; treating W_Assess potential fraud, and W_Shortened completion as exceptions, therefore reducing their occurrences; ensuring *Request Amount* attribute never exceeds $100,000$, and removing 14 resources that are not involved in this new procedure. In light of these changes, the simulation model should maintain the behaviour of the previous procedures.

As a workaround to this issue, the bank could consider defining two separate simulation models and merging their outputs: one for the new procedure, and the existing ones. However, this solution may be impractical when there is insufficient data to build a reliable data-driven simulation model. Another potential solution could be generating an additional number of traces and filtering out the undesired ones a-posteriori. Additionally, both of these workarounds lead to the same issue as of the generative models: global inconsistencies and loss of contextual relevance.

Given the complementary strengths and limitations of generative models and BPS, this work integrates these two types of methods to support complex, realistic what-if analyses.

III. Background

In this section, we introduce the basic concepts required to understand the remainder of the paper.

A. Event Log, Trace, Event

An event log $\mathcal{L}$ records the executions of a business process, i.e., the *traces*. A trace σ_i, in turn, consists of ordered events $\sigma_i = \langle e_{i_1}, e_{i_2}, \dots e_{i_n} \rangle$. Events may be characterized by

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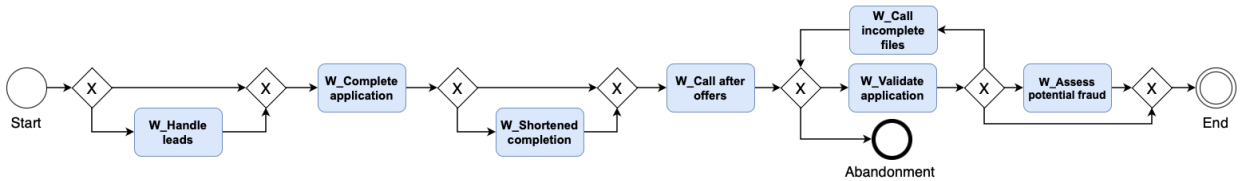

Fig. 1. The BPMN process model of the BPI2017_W event log

multiple attributes (*data attributes*), e.g., an event refers to an activity, may have a timestamp, may be executed or initiated by a given resource, and may have associated other data attributes, i.e., data produced by the activities of the process.

Traces and events can also have *attributes*, also referred to as data payloads. While the values of some attributes may vary throughout the execution of a trace, some attributes remain constant, such as the *Loan Goal* or *Application Type* in Section II. A trace σ_i is represented as:

$$\sigma = \langle (a_1, T_1, \mathbf{d}_1), \ldots, (a_i, T_i, \mathbf{d}_i) \rangle \tag{1}$$

where each a_i represents the activity, T_i is the activity's *timestamp*, and each d_i is the vector of *event's attributes*, also known as the data payload of the activity, containing information about either the case or event's attributes, such as the Resource performing that activity.

B. Generative Models in Process Mining

In Process Mining, generative tasks predominantly focus on the creation of synthetic traces—sequences of events that simulate plausible process executions. Unlike traditional predictive tasks, which forecast future outcomes of ongoing cases, generative models aim to replicate or explore the possible process behaviour space. These models are used for simulation [15], conditional trace generation [7], privacy-preserving data synthesis [14], and more recently, for generating alternative scenarios in counterfactual reasoning [6].

Trace generation refers to the task of synthetically producing sequences (traces) that represent plausible executions of a business process. More formally, given a trace generator function G_θ, the goal is to generate a set of traces $\mathcal{S} = \{\sigma_1, \sigma_2, ..., \sigma_n\}$, where each trace is a sequence of events, composed of the activity labels a_{n_j}, activity timestamp values T_{n_j} and the data payload of the event d_{n_j}, as in (1), structured to adhere to behavioural, temporal, and data-related properties of the original process. These n generated traces in $\mathcal{S}$ should all pertain to the desired condition(s) that was specified apriori.

In data-driven approaches, G_θ is typically instantiated as a neural architecture—such as Recurrent Neural Networks (RNNs), Variational Autoencoders (VAEs), or Generative Adversarial Networks (GANs) [7], [13]–[15]. Alternatively, G_θ can represent a non-parametric or optimization-based method such as Genetic Algorithms, particularly in counterfactual generation scenarios [6]. These methods do not directly model the full trace distribution of all possible traces, but instead optimize for traces satisfying desired properties (e.g., achieving (a) desired outcome(s) while remaining minimally different from a given trace), which can then be leveraged to iteratively generate a set of traces satisfying the desired properties.

C. Business Process Simulation (BPS)

A BPS model is a tuple $\mathcal{M} = (\mathcal{N}, \mathcal{P})$ composed of a business process model $\mathcal{N}$ (e.g., a BPMN model), and a set of simulation parameters $\mathcal{P} = \{\mathcal{P}_R, \mathcal{P}_T, \mathcal{P}_C, \mathcal{P}_S, \mathcal{P}_D\}$. The simulation parameters correspond to the resource, the time, the control flow, the inter-arrival time and the data payload perspective, respectively. In particular, $\mathcal{P}_R$ specifies the resources grouped in roles involved in $\mathcal{M}$ along with allocation rules for activity executions. $\mathcal{P}_T$ defines the processing time required to complete each activity in $\mathcal{N}$. In the presence of decision points in $\mathcal{N}$, $\mathcal{P}_C$ handles the selection among possible paths; otherwise, the execution of a trace follows the semantics of $\mathcal{N}$. Finally, $\mathcal{P}_S$ defines case inter-arrival time, which determines when new traces are instantiated during the simulation, and $\mathcal{P}_D$ defines the data payloads for events and traces.

BPS models are discrete event simulation (DES) where multiple traces are generated simultaneously based on the inter arrival time, and compete for the resource to complete the process. Specifically, an activity within a trace can be executed if at least one resource according to $\mathcal{P}_R$ is available; otherwise, it waits until the necessary resource is available. Moreover, $\mathcal{P}_R$ includes the calendars associated with the resources involved in the process, which define their availability. Hence, in the presence of calendars, a resource can execute an activity if it complies with the predefined rules in $\mathcal{P}_R$ and is available according to its calendar. The waiting times observed during the simulation of the process are caused by resource contention and limited resource availability. The output of the simulation $\mathcal{M}$ is a simulated event log containing the generated traces.

IV. Related Work

Multiple generative Artificial Intelligence (GenAI) techniques have been proposed in the Process Mining literature. Most of them are grounded in Predictive Process Monitoring and generate entire traces by iteratively predicting the next event. The predominant approaches rely on Recurrent Neural Networks (RNNs), such as LSTM [4] and GRU [3]. These techniques have been enhanced through various deep learning strategies, including learnable embeddings for categorical variables [15] and adversarial training [13], [14]. However, these models remain inherently rigid, limited to reproducing behaviours observed in historical data. More recently, the introduction of conditional deep learning generative models for

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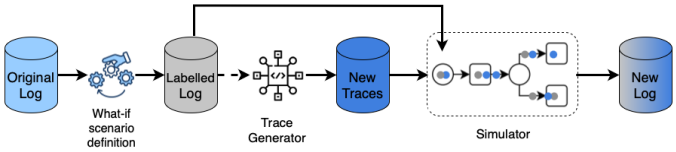

Fig. 2. WISDOM pipeline, outlining the steps of the proposed approach.

trace generation [7], [12] has addressed this issue by enabling some level of control over the generated traces.

In a different direction, counterfactual explanations [6] have been applied as Explainable AI techniques in predictive process monitoring tasks, generating what-if explanations to achieve a desired outcome for a given input trace. As such, they can be leveraged to create what-if scenarios by transforming traces associated with one condition into traces reflecting a different, desired scenario.

On the other hand, Business Process Simulation (BPS) has seen rapid advancements in recent years due to its ability of generating traces in an inter-dependent manner. Recently, hybrid BPS models [15], [16] have been introduced to address the oversimplification and the inability of traditional Discrete Event Simulation (DES) approaches to accurately reproduce the distributions observed in real processes [3], [17]. Thanks to their transparent structure, BPS models are often regarded as a natural tool for conducting what-if scenario analysis [5], especially for simple scenarios regarding resource management [8], [9]. Slightly more complex scenarios, such as the introduction of a previously unseen activity [15] or the need to modify several simulation parameters, as shown in Section II, present significant challenges.

For these reasons, in this work we propose a framework where a BPS model is used to adjust inter-trace dependencies in traces generated by other models—models that may be more effective at defining complex what-if scenarios and producing accurate individual traces, but that lack the ability to capture global process-level interactions. In the literature, only [18] offer a combination of a GenAI method with BPS to generate what-if scenarios. However, the approach is limited to control-flow scenarios and a narrow set of changes within it. In contrast, we propose a general framework that integrates various generative methods across multiple process perspectives.

V. Approach

Figure 2 illustrates the steps of the proposed approach. The first step of WISDOM (What-If Scenarios Data-Driven generative mOdel and siMulation) involves defining a what-if scenario on the original event log by labeling traces to indicate whether they belong to the what-if scenario or not. Then, a trace generator is trained on the resulting labeled event log, and the newly generated traces are simulated alongside (part of) the original ones to refine them and ensure compliance.

A. What-if scenario Definition

Since our approach primarily relies on data-driven generative models capable of reproducing process behaviours present in an event log, defining a what-if scenario requires properly labeling the existing traces. Consequently, what-if behaviour must be present in the log, or deterministically derivable from it, to be used by the model during training. The label can represent a characteristic of the trace and may refer to one or more perspectives of the process, or even to its outcome. For instance, in the loan application process described in Section II, the what-if label identifies traces that correspond to requests for loan limit increases to purchase a car. Formally, we define the what-if label as y and let $\mathcal{L}_y$ be the event log in which each trace is labelled to indicate whether it belongs to the desired scenario ($y = +$) or not ($y = -$). WISDOM can also be applied to a multi-class setting, where each label represents a different what-if scenario. In this way, the model can generate distinct traces corresponding to multiple scenarios simultaneously. For simplicity, in the following section we refer to the binary what-if scenario case.

B. Traces generation

The trace generator G_θ takes as input either the entire log $\mathcal{L}_y$ or a portion of it, depending on the goal of the what-if scenario assessment. For instance, in the loan application process described in Section II, we can assess the effect of the new desired procedure on the entire process by using the full event log. Alternatively, one can focus on a specific subset—such as traces that span the time window of a few months—to analyse its impact over this specific time window in terms of the seasonality behaviour of the process and its congestion.

G_θ models the distribution of event sequences within $\mathcal{L}_y$, capturing the common patterns associated with the classes of the label y, by learning temporal dependencies, activity transitions, and relevant contextual features present in the traces. After training, the model G_θ is used to generate new traces $\hat{\sigma} \in \mathcal{L}_{\text{gen}}$ conforming to the expected behaviour of the class $y = +$, i.e., traces that satisfy the what-if scenario.

We highlight the fact that WISDOM is agnostic with respect to the choice of the traces generator, meaning that it can accommodate various methods for generating synthetic traces, such as deep learning models (e.g., RNNs, LSTMs), probabilistic models, or rule-based systems, depending on the available data and specific needs of the scenario being analysed. Moreover, the trace labelling step depends only on the types of generators that require it. WISDOM is also capable of integrating unseen process behaviours.

C. Simulation Replay

To ensure the conformance of the generated traces to the process, particularly regarding the time and resource perspective, we rely on a Business Process Simulation (BPS) technique. The idea is to leverage simulation to replay the traces $\mathcal{L}_{\text{gen}}$ generated by G_θ together with the original labeled log $\mathcal{L}_y$, or at least the part of it that is still relevant for the what-if scenario considered. The use of BPS allows proper handling of inter-trace resource contentions, ensuring consistent waiting times across co-executing traces. In particular, inter-trace dependencies arise when activities from different

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traces simultaneously require the same resource. Omitting these contentions can easily lead to inaccurate waiting times.

For the BPS component, we opted for hybrid simulation techniques [1], given their state-of-the-art performance and their ability to perform trace replay [16], which is required by our method. Specifically, during trace replay, we leverage the control-flow perspective of GenAI models, given their superior performance compared to traditional BPS models, as shown in [16]. Accordingly, we define the BPS model $\mathcal{M}^\theta = (\mathcal{L}_y \cup \mathcal{L}_{\text{gen}}, \mathcal{P}^\theta)$, where $\mathcal{L}_y$ and $\mathcal{L}_{\text{gen}}$ represent the traces to be replayed — drawn from the original and generated logs, respectively — and $\mathcal{P}^\theta$ denotes the simulation parameters used to manage their execution. To correctly replay traces in $\mathcal{L}_y$ and $\mathcal{L}_{\text{gen}}$ without a process model, we represent them as partial orders to capture any potential parallelism between activities. The traces and event attributes are also extracted from $\mathcal{L}_{\text{gen}}$ and $\mathcal{L}_y$ and are recorded as sequences in $\mathcal{P}_D^\theta$. The parameters $\mathcal{P}_T^\theta$, $\mathcal{P}_S^\theta$, and $\mathcal{P}_R^\theta$ are defined to allocate resources, grouped by roles together with their specific calendars. Specifically, $\mathcal{P}_R^\theta$ preserves role assignments for each trace σ in $\mathcal{L}_y$, while for generated traces $\hat{\sigma}$ in $\mathcal{L}_{\text{gen}}$, roles are inferred based on patterns learned from $\mathcal{L}$. $\mathcal{P}_T^\theta$ is used to estimate processing times only for the activities in $\hat{\sigma}$, whereas the corresponding times in σ remain unchanged. Finally, $\mathcal{P}_S^\theta$ is defined as a distribution function that ensures traces are spread across the process timeline, preventing unrealistic stacking of them. In this way, we can properly assess the effect of $\mathcal{L}_{\text{gen}}$ on $\mathcal{L}_y$, which preserves the time patterns—apart from the waiting times, which are recomputed due to a different level of process congestion—of the original traces, while estimating those of the new traces. Consequently, the BPS handles inter-trace dependencies by processing traces in parallel based on their start times. By replaying the traces concurrently, resource contention impacts time and other perspectives, as traces have to wait for resource availability, but also ensures compliance across correlated perspectives like time and calendars. In the end, the BPS returns $\mathcal{L}_{\text{sim}}$ as the integration of $\mathcal{L}_y$ and $\mathcal{L}_{\text{gen}}$.

VI. Evaluation

In this section, we present the evaluation methodology used to assess the quality of what-if scenarios generated using both standalone generative models and their WISDOM integration [2]. We aim to answer the following research questions:

RQ1 What are the trade-offs between GenAI methods and BPS methods in generating realistic what-if scenarios from control-flow and resource perspectives?

RQ2 How do the generated logs obtained with WISDOM using different trace generators compare to what-if scenarios created entirely using BPS methods?

In answering **RQ1**, we will show GenAI models' ability to produce event logs that closely follow control-flow constraints at the individual trace level, that however fail to respect resource-related dependencies when considering the set of traces as a whole. In contrast, BPS approaches can preserve resource realism across the full set of traces, while the individual traces lag behind in quality. To answer **RQ2**, we focus on the integration of generative methods within WISDOM to address the aforementioned limitations and compare them against a pure BPS technique. To support this, we develop an evaluation pipeline inspired by the methodology proposed in [15], [18]. Specifically, we modify the original log by removing a specific part of the process, which then defines the target what-if scenario (see Figure 3). This allows the direct comparison of the generated what-if traces with the original log, assessing the realism and quality of the generated scenarios.

As BPS simulator in WISDOM we use RIMS [16], given its flexibility in being used as a trace replayer. To properly compare WISDOM and the purely BPS-based technique in **RQ2**, we employ the same simulation parameters, $\mathcal{P}^\theta$.

TABLE I
Datasets Description used in the evaluation

Dataset	Trace #	Variant #	Activity #	Avg. trace length	Avg. trace cycle time
Sepsis	782	733	16	17	38.2d
BPIC2012	4685	3790	36	40	18.7d
BPIC2017_W	30276	4206	8	6	12.5d

A. Datasets

The evaluation spans over 3 real-life event logs, where a what-if scenario is defined for each one. In order to replay the traces as a step of WISDOM and to properly account for the resource allocation and the resulting congestion, it is necessary to consider the start timestamps of activities. Therefore, we consider datasets that already contain both start and end timestamps for each activity (*BPIC2017_W*) or we preprocess the logs by estimating the start timestamps (*BPIC2012*, *Sepsis*) using the approach proposed in [19].[3]

- **Sepsis cases** [20] contains events of sepsis cases from a hospital. We consider the following what-if scenario: an increase in the number of patients under the age of 80, taking into account their specific clinical and treatment patterns in the hospital.
- **BPIC2012** [21] contains a loan application process. We consider the following what-if scenario: an increase in the number of traces where the loan offer is accepted.
- **BPIC2017_W** [11] contains a similar loan application process as **BPIC2012**, as describes in Section II. We consider the following what-if scenario: an increase of loans requests with "Loan Goal": *Home improvement* or *Car* and "Application Type": *New Credit*.

Table I reports details on the three datasets in terms of number of traces, activity labels and cycle times. The what-if scenarios focus on trace attributes in Sepsis and BPIC2017_W, while BPIC2012 emphasizes the control-flow perspective.

[1]Trace alignment techniques require an explicit process model for trace replay and can only handle the control-flow perspective.

[2]WISDOM repository: https://github.com/rgraziosi-fbk/wisdom.

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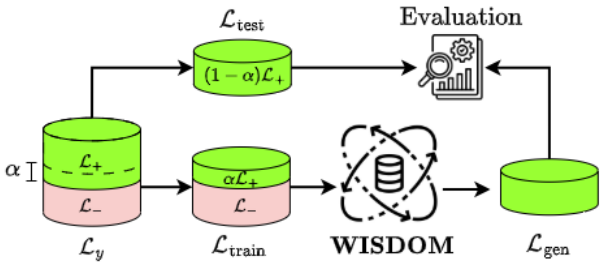

Fig. 3. Evaluation pipeline: the labelled log $\mathcal{L}_y$ is split into $\mathcal{L}_{\text{train}}$ which is used as input for WISDOM to generate the what-if scenario traces of $\mathcal{L}_{\text{gen}}$, which is then compared to $\mathcal{L}_{\text{test}}$

B. Experimental settings

Figure 3 illustrates the experimental setup designed for evaluating the what-if scenario generation. Based on the label y, the log $\mathcal{L}_y$ is partitioned into two disjoint subsets: the positive traces $\mathcal{L}_+ = \{\sigma_i \in \mathcal{L}_y \mid y_i = +\}$, which satisfy the what-if scenario condition, and the negative traces $\mathcal{L}_- = \{\sigma_i \in \mathcal{L}_y \mid y_i = -\}$, which do not.

We introduce a parameter $\alpha \in [0, 1]$, called the *sampling factor*, to control the proportion of traces from the what-if scenario class included in the training set. We construct the training set $\mathcal{L}_{\text{train}}$ by combining all traces from the class $\mathcal{L}_-$ with a random sample $\mathcal{L}_\alpha \subset \mathcal{L}_+$ of size $|\mathcal{L}_\alpha| = \alpha \cdot |\mathcal{L}_+|$ from the what-if scenario class $\mathcal{L}_+$. We define the training set as $\mathcal{L}_{\text{train}} = \mathcal{L}_- \cup \mathcal{L}_\alpha$, while the remaining traces from the what-if scenario class form the test set $\mathcal{L}_{\text{test}} = \mathcal{L}_+ \setminus \mathcal{L}_\alpha$. This reflects a realistic use case where process instances are generated under a hypothetical condition, controlled by varying levels of α.

As the trace generator G_θ in WISDOM, we instantiate three state-of-the-art techniques, each representing a different generative paradigm, chosen based on their performance on event data generation tasks. First, we use a Conditional Variational Autoencoder (CVAE), which conditions the generation process on the target class label to synthesize traces that reflect the desired what-if condition. We train the CVAE model proposed in [12] using $\mathcal{L}_{\text{train}}$. Second, we implement a Counterfactual Generation (CF) approach. We first train a predictive model p_m as an "oracle" function on $\mathcal{L}_{\text{train}}$. Then, p_m is used to instantiate a Genetic Algorithm approach, as proposed in [6], which modify traces from $\mathcal{L}_{\text{train}}$ that belong to the opposite class ($\mathcal{L}_-$) of the desired what-if scenario, flipping their predictions to achieve the target condition. Lastly, we include a BPS-based approach in two ways: (i) as a third generative funtion G_θ for WISDOM, and (ii) as a purely BPS method. For this, we adopt SIMOD [22], a state-of-the-art BPS simulator and the only one that supports attribute generation [23].[4] We refer to the employed approaches as $\text{WISDOM}_{\text{CVAE}}$, $\text{WISDOM}_{\text{CF}}$, $\text{WISDOM}_{\text{SI}}$ and SIMOD.

[3]For *BPIC2012* and *Sepsis* logs, we also filtered out incomplete traces.

[4]We use two BPS tools for WISDOM and for the evaluation, as RIMS cannot generate attributes, while SIMOD does not allow for trace replay.

For the first two trace generators, we generate the same number of traces as in $\mathcal{L}_{\text{test}}$, as they inherently satisfy the condition expressed as $y = +$. In contrast, with SIMOD, we need to generate more traces than the size of $\mathcal{L}_{\text{test}}$ to ensure a sufficient number of traces respecting $y = +$, as shown in Figure II. This is because defining complex what-if scenarios as defined in Section II, requires adjusting numerous simulation parameters, making the approach impractical in this context. As motivated in Section II, creating a dedicated simulation model specifically for the what-if scenario is unfeasible due to possbile small samples (low α), and neglects the influence of $\mathcal{L}_-$ on the overall process.

C. Evaluation metrics

To answer **RQ1**, we employ the *Conformance score* and *Overlap* metrics. *Conformance score* evaluates how well individual traces, generated by the G_θ methods, complies with the DECLARE [24] constraints discovered from $\mathcal{L}$ with support of 90%, as proposed in [6], [7]. *Overlap* measures resource overutilization in terms of hours, that is, the total number of hours in $\mathcal{L}_{\text{gen}}$ during which resources perform activities that exceed their capacity, discovered from $\mathcal{L}_{\text{train}}$.

In the second phase of the evaluation (**RQ2**), we compare the logs resulting from the different choices of G_θ within WISDOM, as well as the one obtained with vanilla simulation SIMOD, with $\mathcal{L}_{\text{test}}$. To assess the quality of the generated traces, a subset of the metrics introduced in [10] within the domain of simulation models is used. These metrics quantify the "distances" between the simulated traces and those in the $\mathcal{L}_{\text{test}}$ by computing the Earth Mover's Distance (EMD) of their respective distributions along different temporal and control-flow dimensions. For the control-flow evaluation, we leverage *Control-Flow Log Distance* (CFLD), which minimizes the total distance by pairing the traces from the generated and test logs using the Hungarian algorithm [25]. For the temporal dimension we employ two metrics: *Cycle Time Distribution* (CTD), measuring the distance between the cycle time distributions of all traces, and *Relative Event Distribution* (RED), analysing the temporal distribution of events relative to the case origin (i.e., the case arrival). Finally, for the resource perspective, we use the *Circadian Workforce Distribution* (CWD), which evaluates the distribution of active resource allocations across the days of the week within the process. For all metrics—except for CONF, where higher is better—a lower value indicates that the simulated log is closer to $\mathcal{L}_{\text{test}}$.

VII. RESULTS

In this section, we aim to answer the research questions outlined in Section VI. We answer **RQ1** by evaluating the different trace generators G_θ (i.e., CVAE, CF, SIMOD), and then proceed to **RQ2** where we compare the WISDOM variants against the purely BPS-based approach, SIMOD.

a) Answering RQ1: Table II reports the compliance in terms of control-flow and resource perspective of G_θ methods with respect to the sampling factor (α) in terms of the conformance score with respect to the DECLARE constraints,

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TABLE II
COMPARISON OF TRACE GENERATORS ACROSS VARYING SAMPLING FACTORS (α) FOR THE SEPSIS DATASET

G_θ	α	Conformance (%)	Overlap	# traces gen.
CF	0.3	99%	1431 hrs	168
	0.2	98%	7965 hrs	192
	0.15	96%	6242 hrs	204
	0.1	96%	10113 hrs	216
	0.05	99%	11920 hrs	228
CVAE	0.3	94%	6134 hrs	168
	0.2	97%	6163 hrs	192
	0.15	96%	41466 hrs	204
	0.1	96%	15232 hrs	216
	0.05	96%	39712 hrs	228
SIMOD	0.3	87%	0 hrs	1161
	0.2	88%	0 hrs	2462
	0.15	88%	0 hrs	2291
	0.1	85%	0 hrs	955
	0.05	87%	0 hrs	6853

the resource overlap expressed in hours, and the number of generated traces required to reach the size of $\mathcal{L}_{test}$ [5].

When looking at the Overlap scores across the different α, a clear pattern emerges: the GenAI methods (i.e. CVAE and CF) register a large overlap score, which increases as α decreases (i.e. as the dimensionality of $\mathcal{L}_{test}$ increases). In terms of conformance, both CVAE and CF perform well in generating traces that respect the constraints of the underlying process, confirming the fact they can indeed generate a higher number of conformant control-flow executions when compared to BPS approach. For the other two event logs, we can observe that in Table III the same behaviour is observed as SIMOD tends to perform worse for BPIC2012 and exhibits great results in BPIC2017_W. Finally, as mentioned in Section VI-B, SIMOD requires the generation of a larger number of traces to produce the required length of $\mathcal{L}_{gen}$, varying from an increase of 4-fold (when $\alpha = 0.1$) to 30-fold (when $\alpha = 0.05$), as opposed to the CVAE and CF methods, which can directly generate the number of desired traces pertaining to $y = +$.

Ultimately, this analysis confirms the hypothesis that while GenAI models can produce conformant traces more effectively than BPS, they are not able to ensure that the generated traces consistently reflect the desired what-if scenario while also preserving inter-trace dependencies, such as resource overlap. Overall, for **RQ1**, between the different trace generators, CF strikes the best balance among the three methods, combining high conformance and moderate resource overlap. CVAE excels in control-flow conformance but struggles with the overlap error in most of the cases. While SIMOD is able to ensure that no overlap occurs in $\mathcal{L}_{gen}$, it exhibits inferior conformance scores and requires the generation of a potentially extremely large number of traces to reach the desired what-if scenario. As a result, we integrate the three generators G_θ in WISDOM, solving the resource overlap issue of the GenAI methods.

[5]We detail in Table II the results corresponding only to the Sepsis dataset, the results for the other datasets are qualitatively similar and are reported on the supplementary material within the repository.

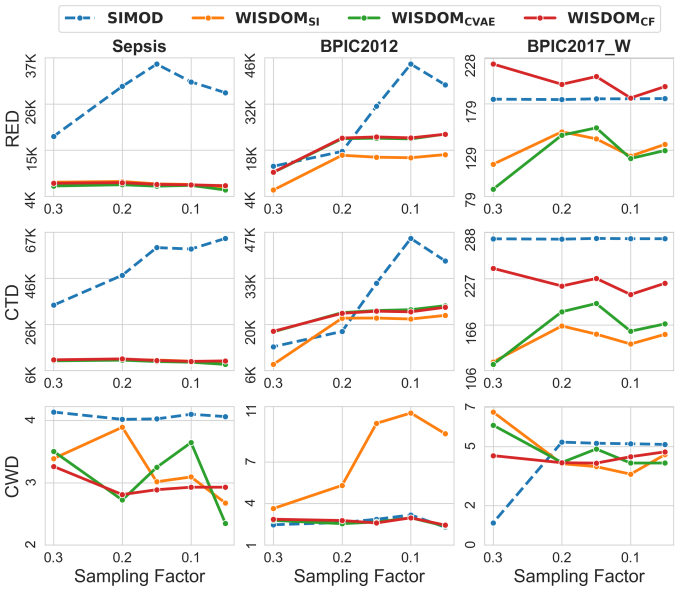

Fig. 4. Time-related evaluation metrics for each method and dataset across varying levels of the sampling factor α

TABLE III
CONTROL-FLOW RELATED METRICS FOR EACH DATASET AND METHOD

Dataset	Method	CFLD	NGD	Conformance (%)
Sepsis	SIMOD	0.4002	0.5864	87%
	WISDOM$_{SI}$	0.4493	0.6720	85%
	WISDOM$_{CVAE}$	0.4224	0.5389	92%
	WISDOM$_{CF}$	0.4352	0.5489	92%
BPIC2012	SIMOD	0.5434	0.6388	61%
	WISDOM$_{SI}$	0.7873	0.8217	25%
	WISDOM$_{CVAE}$	0.4352	0.6412	83%
	WISDOM$_{CF}$	0.5292	0.7460	91%
BPIC2017_W	SIMOD	0.3131	0.3609	100%
	WISDOM$_{SI}$	0.3098	0.3089	100%
	WISDOM$_{CVAE}$	0.0657	0.0546	95%
	WISDOM$_{CF}$	0.6856	0.8353	100%

b) Answering RQ2: Table III presents the performance of WISDOM and SIMOD concerning the control-flow perspective of the generated logs, averaged across sampling factors, as no major differences were observed. As expected, the integration with WISDOM does not significantly impact the *Conformance* score, with the only exception being *BPIC2012*. For this log, WISDOM$_{SI}$ reduces performance by half compared to the vanilla SIMOD in terms of *Conformance* score, which in turn leads to an increase in CFLD and NGD metrics. This is due to they way the parallelisms are captured for *BPIC2012*, as SIMOD employs a process model while WISDOM uses the partial orders between activities, thus introducing a decrease in conformance in the WISDOM$_{SI}$ integration. Nevertheless, WISDOM$_{CVAE}$ and WISDOM$_{CF}$ ensure a high conformance score and show similar performance on the CFLD and NGD metrics compared to SIMOD. The effect of parallelism is also evident in Figure 4, which reports the metrics related to the time and resource perspective, where it impacts the distribution of resources over the weekdays (CWD metrics), especially for WISDOM$_{SI}$, leading to greater error compared to SIMOD and the other two WISDOM variants. Overall,

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WISDOM and its variants outperform SIMOD, with the only exception being $\text{WISDOM}_{\text{SI}}$ in BPIC2012, which is impacted by parallelism, as previously discussed. Moreover, WISDOM shows robustness with respect to α, especially for the RED and CTD metrics, showing overall stable results. In contrast, SIMOD struggles with data scarcity (low α), particularly in the Sepsis and BPIC2012 logs, resulting in higher CTD and RED values. Meanwhile, BPIC2017_W remains more stable, as the larger log size (Table I) provides a sufficient number of traces to train the simulation model effectively.

For **RQ2**, we showed that the integration of trace generators in WISDOM, especially GenAI techniques, guarantees compliance to inter-trace dependencies and improves the overall stability across different levels of the sampling factor (α). SIMOD also performs well under certain conditions, although it struggles with data scarcity reflected in the poor results for the time and resource perspectives, as shown in Fig. 4.

VIII. Conclusions

In this work, inspired by the limitations of GenAI models and BPS techniques in generating consistent event logs for complex what-if scenarios, we proposed WISDOM as a method to combine the two families of techniques. With the capacity of GenAI techniques in generating complex what-if scenarios and BPS's capabilities of ensuring the consistency of all generated traces, WISDOM can ensure realistic what-if scenario generation across multiple process perspectives. In our evaluation, we tested two state-of-the-art GenAI techniques and compared them against a BPS technique. We then integrated all three methods with WISDOM, and compared them against the BPS method. The results show that the WISDOM variants leveraging GenAI models outperform the BPS approach in most settings—particularly with respect to resource allocation and the timing aspect—whilst preserving high control-flow conformance.

A limitation of WISDOM when using GenAI techniques is its reliance on existing log data to define what-if scenarios, which however enables ground-truth comparisons. The generation of entirely novel what-if scenarios is typically knowledge-driven and relies on BPS simulators, which can be integrated in WISDOM, as shown in Section VI. However, as discussed earlier in this paper, defining complex scenarios through BPS parameter setting remains a challenging task, particularly when these parameters span multiple process perspectives. However, this would require very careful evaluation and is left for future work.In this study, we focused on binary what-if scenario conditions. As future work, we plan to investigate multi-class scenarios and evaluate alternative generator approaches. Since WISDOM supports multi-class conditions, it easily simulates traces associated with various scenarios, as the trace generators are capable of handling multiple conditions as well. Due to its modular design, WISDOM allows seamless integration of any trace generator or BPS technique that supports trace replay.

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